JORGE J. ORTIZ

AI Evaluation & Research Leader | Trustworthy Agentic Systems

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Research Leadership Profile

Hands-on AI research leader who builds evaluation systems for models and agents operating under real-world uncertainty. Research connects long-horizon agent evaluation, formal verification, runtime monitoring, multimodal capability measurement, and explicit abstention. The central question is not only what a system can do, but when its evidence is sufficient to act. Director of the Sensing & Reasoning Lab; Rutgers site director or lead for two NSF centers; Associate Professor with tenure; and applied AI lead in professional sports. Experienced setting research direction, mentoring researchers and engineers, writing production-oriented research code, and communicating limitations and evidence to technical and non-technical stakeholders.

Relevant Research & Evaluation Leadership

Director, Sensing & Reasoning Lab — **2018–Present**
Rutgers University — *Piscataway, NJ*

- Set the research agenda for trustworthy agentic AI, multimodal reasoning, causal inference, and self-assessing systems; lead projects from hypothesis and experimental design through implementation, deployment, publication, and external communication.
- Mentor multidisciplinary researchers and engineers across computer science, engineering, human–robot interaction, health, and urban systems. Graduated three Ph.D. students, with alumni now at Meta and Colby College. Two current Ph.D. students expected to graduate in Fall 2026 are set to join Apple AI and Waymo.
- Build evaluation methods around explicit competency boundaries, calibrated uncertainty, failure analysis, and human escalation instead of benchmark score alone.

Rutgers Site Director, Center for Responsible AI & Governance (CRAIG) — **2025–Present**
NSF Industry–University Cooperative Research Center — *Rutgers University*

- Lead a cross-sector research program on the evidence required to justify AI deployment, translating technical evaluation results into reusable governance and decision frameworks.
- Develop evaluation pipelines for agentic systems that combine formal guarantees, runtime evidence, guardrails, and auditable post-training behavior.

Rutgers Site Lead, Center for Smart Streetscapes (CS3) — **2022–Present**
NSF Engineering Research Center — *Rutgers University*

- Lead Rutgers research within a $26M multi-institution center; align sensing, perception, privacy, and causal reasoning work across academic, government, community, and industry partners.
- Drive evaluation in deployed settings where distribution shift, incomplete evidence, and operational constraints make capability boundaries consequential.

Selected Work in Model Capabilities, Agents & Measurement

Veritas for Epistemic Oversight of Long-Running AI Agents — **2026–Present**
Research system and multi-model evaluation study under preparation for submission

- Veritas tests whether long-running AI agents can support their claims with evidence and recognize when they should proceed, revise their work, or defer to a human.
- The system turns claims into explicit verification obligations, runs independent checks, and preserves auditable traces and evidence bundles for each decision.
- My group and I are preparing this work for submission. The study uses matched conditions across multiple model families, seeded faults, clean-task do-no-harm panels, independent scoring, and predeclared budgets and stopping rules.

TraceFix for Verified Coordination in Agentic Systems — **ACM CAIS 2026**
Outstanding Solution Paper

- Use TLA+ model checking and counterexample-guided repair to find and fix deadlocks, races, and protocol violations in multi-agent coordination; enforce verified protocols with runtime topology monitoring.

Explicit Abstention for Predictable VLM Reliability — **2026**
Preprint

- Developed explicit abstention controls for video question answering to expose capability–coverage tradeoffs and make system reliability more predictable under uncertainty.

Human-Centered RLHF for Autonomous Driving — **IMWUT/UbiComp 2024**
Model behavior and safety

- Co-developed an LLM-enhanced RLHF approach that incorporates human-centered safety preferences into autonomous-driving optimization and evaluation.

Professional Experience

Associate Professor (with Tenure), Electrical & Computer Engineering **2025–Present**
Rutgers University *Piscataway, NJ*

- Principal investigator or co-investigator on a $32.4M portfolio across responsible AI, urban AI, health sensing, and intelligent infrastructure; balance scientific rigor with milestones, partner needs, and deployed-system constraints.
- Leadership Research Faculty Fellow (2026), developing best practices for thoughtful AI use in research development.

Assistant Professor, Electrical & Computer Engineering **2018–2025**
Rutgers University *Piscataway, NJ*

AI & Computer Vision Lead, Baseball Operations **2019–Present**
New York Yankees *New York, NY*

- Lead applied computer-vision and biomechanics research for player development and performance, with an emphasis on robust measurement and signaling uncertainty in high-stakes decisions.

Research Staff Member **2013–2018**
IBM Research *Yorktown Heights, NY*

- Researched and built machine-learning, sensing, and distributed systems that moved from experimental prototypes toward real operating environments.

Senior Software Engineer / Software Engineer **2003–2013**
Spire Global / Oracle *San Francisco Bay Area*

Selected Publications & Research Signals

- **Xia, Li, Ehsan, Ortiz.** “TraceFix, Repairing Agent Coordination Protocols with TLA+ Counterexamples.” *ACM Conference on AI and Society*, 2026. Outstanding Solution Paper.
- **Ortiz.** “Explicit Abstention Knobs for Predictable Reliability in Video Question Answering.” arXiv 2601.00138, 2026.
- **Sun, Contreras, Ortiz.** “DFGauss, Dynamic Focused Masking for Autoregressive 3D Occupancy Prediction.” *NeurIPS*, 2025.
- **Bu, Tsai, Tjokro, Bhattacharjee, Ortiz, Ju.** “Using Vision-Language Models as Proxies for Social Intelligence in Human–Robot Interaction.” 2025.
- Research has appeared across NeurIPS, CoRL, IMWUT/UbiComp, IPSN, SenSys, IoTDI, and USENIX ATC; recipient of multiple best-paper and best-paper-finalist recognitions.

Technical & Organizational Strengths

Evaluation	Long-horizon agents; capability and coverage measurement; fault injection; matched experiments; independent scoring; ablations; negative-result analysis; reproducible evidence
Trustworthy AI	Formal methods (TLA+); runtime monitoring; verification plans; evidence provenance; abstention; uncertainty calibration; causal reasoning; privacy-aware sensing
Research leadership	Strategy and roadmaps; team mentorship; cross-functional collaboration; multi-institution programs; model-release-style technical narratives; stakeholder communication
Implementation	Python, Go, C/C++; evaluation harnesses; ML/CV systems; distributed systems; experiment automation; open-source research artifacts

Education

Ph.D., Computer Science, University of California, Berkeley 2013
M.S., Computer Science, University of California, Berkeley 2010
B.S., Computer Science, Massachusetts Institute of Technology 2003